

Estimating Differences and Checking Answers

Answer Key

Grades 2–3

Quick Answers

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|---|---|
| 1. the estimate = 40 apples | 7. the estimate = 30 marbles |
| 2. the estimate = 30 seats | 8. (a) the estimate = 300, (c) the exact difference = 338, — |
| 3. the estimate = 300 books | 9. the estimate = 500 tickets |
| 4. (a) the estimate = 300, (b) the exact number left = 337, — | 10. (a) the nearest-ten estimate = 220, (b) the nearest-hundred estimate = 200, (c) the exact difference = 224, — |
| 5. the estimate = 500 seats | |
| 6. (a) the estimate = 400, (b) the exact number = 377 | |
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What is verified

15 of 18 answers machine-checked against SymPy, across 10 problems.
 — marks an answer only you can judge — problems 4, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 2–3

3.NBT.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 18 tagged checks.

Common wrong answers

1. *If they got 30: rounded both numbers down instead of to the nearest ten*
 3. *If they got 200: rounded both numbers down instead of to the nearest hundred*
 4. *If they got 437: opened a hundred for the tens but left the hundreds digit unchanged*
 8. *If they got 438: opened a hundred for the tens but left the hundreds digit unchanged*
 9. *If they got 400: rounded the number being taken away up instead of down*
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1. A shop had 78 apples and sold 43. About how many are left, rounding each to the nearest ten?

Solution: Round both, then subtract. The ones digit of 78 is eight, so it rounds up to eighty; the ones digit of 43 is three, so it rounds down to forty. Eighty take away forty leaves the estimate.

40 apples

2. A bus has 92 seats and 57 people are seated. About how many are free, to the nearest ten?

Solution: 92 rounds down to ninety and 57 rounds up to sixty. Ninety take away sixty leaves the estimate. Rounding the second number up makes the estimate a little low, which is the safe direction when you are asking whether there is room.

30 seats

3. A library owns 486 books with 218 on loan. About how many are on the shelves, to the nearest hundred?

Solution: For hundreds, look at the tens digit. 486 has eight tens, so it rounds up to five hundred; 218 has one ten, so it rounds down to two hundred. Five hundred take away two hundred leaves the estimate.

300 books

4. A farm collected 634 eggs and packed 297. Mia says 437 are left. (a) Estimate to the nearest hundred. (b) Find the exact number. (c) Is Mia's answer reasonable?

Solution (a): 634 rounds down to six hundred and 297 rounds up to three hundred, so the estimate is six hundred take away three hundred.

300

Solution (b): Exactly: the ones need help, so regroup twice — five hundreds, twelve tens, fourteen ones. Ones: fourteen take seven leaves seven. Tens: twelve take nine leaves three. Hundreds: five take two leaves three.

337

Solution (c) — instructor-judged. A correct response rejects the claim and uses the estimate to do it. Full credit for the reasoning that rounding says about three hundred eggs should be left, while 437 sits more than a hundred above that — far too big a gap for rounding alone, and exactly the size of error a lost regrouping makes. Accept the exact answer used as supporting evidence, as long as the estimate is named as what raised the doubt. Calling the claim wrong without reference to the estimate earns half credit. Accepting the claim earns no credit.

5. A stadium has 852 seats and 419 were sold. About how many were unsold, to the nearest hundred?

Solution: 852 has five tens, so it rounds up to nine hundred; 419 has one ten, so it rounds down to four hundred. Nine hundred take away four hundred leaves the estimate.

500 seats

6. A school printed 725 flyers and handed out 348. (a) Estimate to the nearest hundred. (b) Find the exact number left.

Solution (a): 725 rounds down to seven hundred (two tens) and 348 rounds down to three hundred (four tens). Seven hundred take away three hundred.

400

Solution (b): Exactly: regroup for the ones — six hundreds, eleven tens, fifteen ones. Ones: fifteen take eight leaves seven. Tens: eleven take four leaves seven. Hundreds: six take three leaves three. The exact answer sits a little below the estimate, because the number taken away was rounded down.

377

7. A jar held 64 marbles and 28 were taken. About how many are left, to the nearest ten?

Solution: 64 rounds down to sixty and 28 rounds up to thirty. Sixty take away thirty leaves the estimate.

30 marbles

8. A warehouse held 803 boxes and shipped 465. Ben wrote 438. (a) Estimate to the nearest hundred. (b) Explain why 438 cannot be right. (c) Find the exact number.

Solution (a): 803 rounds down to eight hundred and 465 rounds up to five hundred, so the estimate is eight hundred take away five hundred.

300

Solution (b) — instructor-judged. A correct response compares the written answer with the estimate and says how big the gap is. Full credit for the reasoning that the estimate is about three hundred, while 438 is more than a hundred above it — rounding can shift an estimate by a few tens, never by a whole hundred, so a regrouping must have gone missing. Accept an answer that names the fault directly: the hundred was opened for the empty tens column but the hundreds digit was never reduced. Saying only that 438 looks too big, without using the estimate, earns half credit. Recomputing the difference without saying what the estimate showed earns no credit.

Solution (c): The ones need help and the tens are empty, so open a hundred: seven hundreds and ten tens, then nine tens and thirteen ones. Ones: thirteen take five leaves eight. Tens: nine take six leaves three. Hundreds: seven take four leaves three. 338

9. A cinema sold 906 tickets over a weekend, 448 of them on Saturday. About how many on Sunday, to the nearest hundred?

Solution: 906 has zero tens, so it rounds down to nine hundred; 448 has four tens, so it rounds down to four hundred. Nine hundred take away four hundred leaves the estimate.

500 tickets

10. Two students estimated $512 - 288$, one to the nearest ten and one to the nearest hundred. (a) Nearest-ten estimate. (b) Nearest-hundred estimate. (c) Exact difference. (d) Which estimate is closer, and why?

Solution (a): To the nearest ten, 512 becomes five hundred and ten, and 288 becomes two hundred and ninety. Subtracting those gives the first estimate. 220

Solution (b): To the nearest hundred, 512 becomes five hundred and 288 becomes three hundred. Subtracting those gives the second estimate. 200

Solution (c): Exactly: regroup for the ones — four hundreds, ten tens, twelve ones. Ones: twelve take eight leaves four. Tens: ten take eight leaves two. Hundreds: four take two leaves two. 224

Solution (d) — instructor-judged. A correct response picks the nearest-ten estimate and explains the size of the rounding. Full credit for the idea that rounding to tens moves each number by only a few, while rounding to hundreds can move it by dozens, so less is thrown away and the estimate stays nearer the true answer. Accept an answer that measures both gaps and uses those numbers as the reason. Choosing correctly with no reason earns half credit. Choosing the nearest-hundred estimate, or arguing that a rounder-looking number must be more accurate, earns no credit.